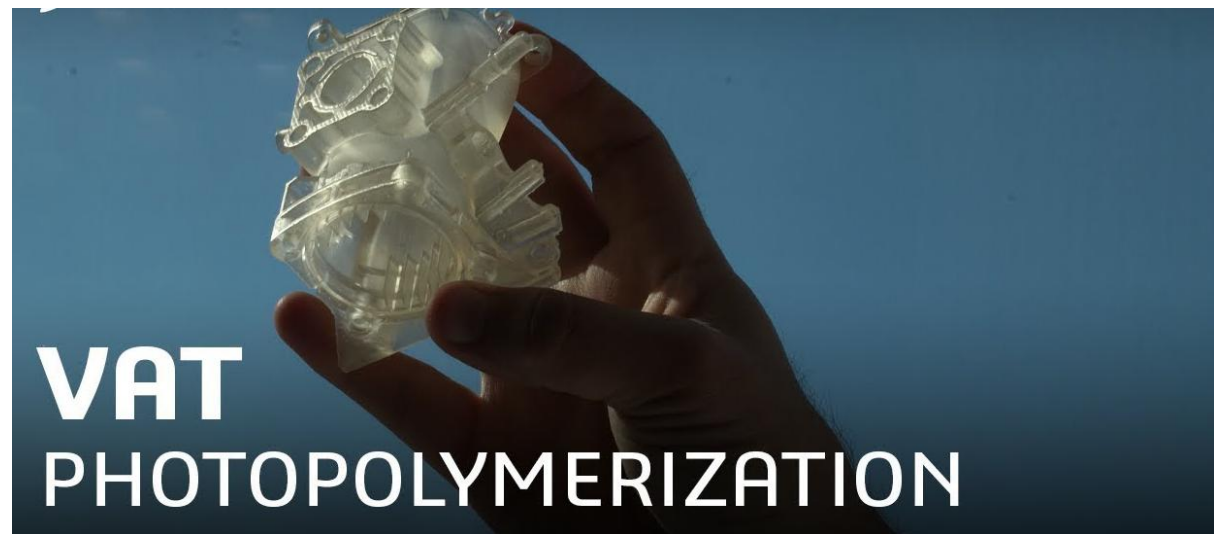




COURSE CONTENT

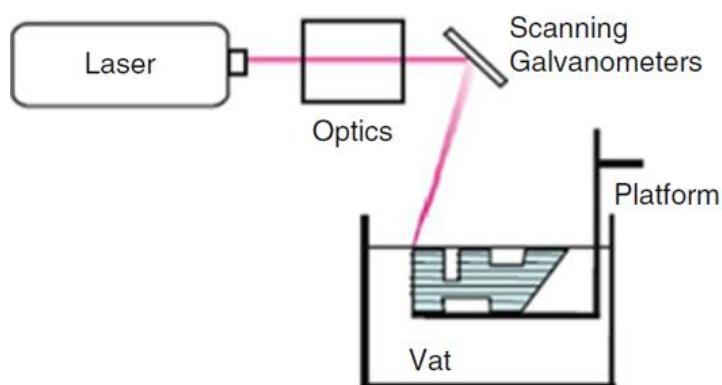
- 1.2.1 Introduction**
- 1.2.2 Stereolithography Apparatus (SLA)**
- 1.2.3 Applications of SLA**
- 1.2.4 SLA Materials**

Prof. Shiva Krishna J

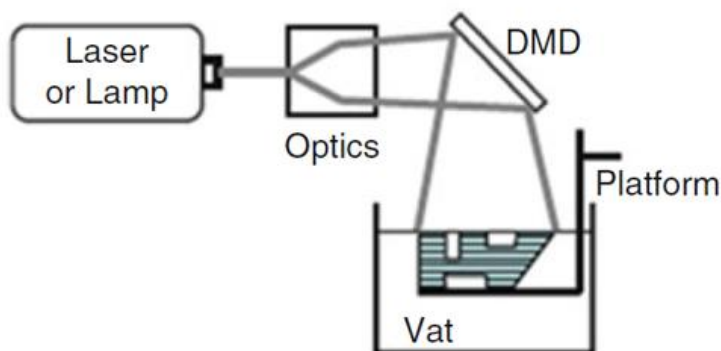
Dept. of Mechanical Engg., Agnel Institute of Technology and Design, Assagao

1.2.1 INTRODUCTION

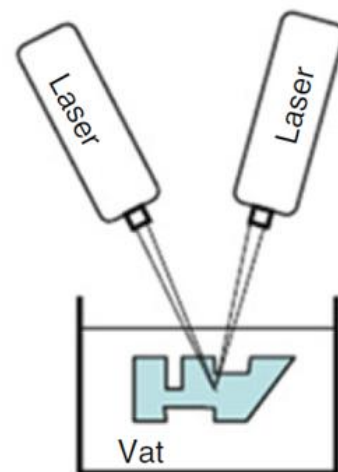
- VAT Polymerization (or) Photopolymerization processes make use of liquid, radiation-curable resins, or photopolymers, as their primary materials.
- Most photopolymers react to radiation in the ultraviolet (UV) range of wavelengths, but some visible light systems are used as well.
- Upon irradiation, these materials undergo a chemical reaction to become solid.
- The VAT Polymerization configurations include:
 1. Vector scan, or point-wise, approaches typical of commercial SL machines
 2. Mask projection, or layer-wise, approaches, that irradiate entire layers at one time,
 3. Two-photon approaches that are essentially high-resolution point-by-point approaches



Vector scan approach



Layer-wise, approach



Two-photon approach

The following process make use of VAT polymerization technique for building parts:

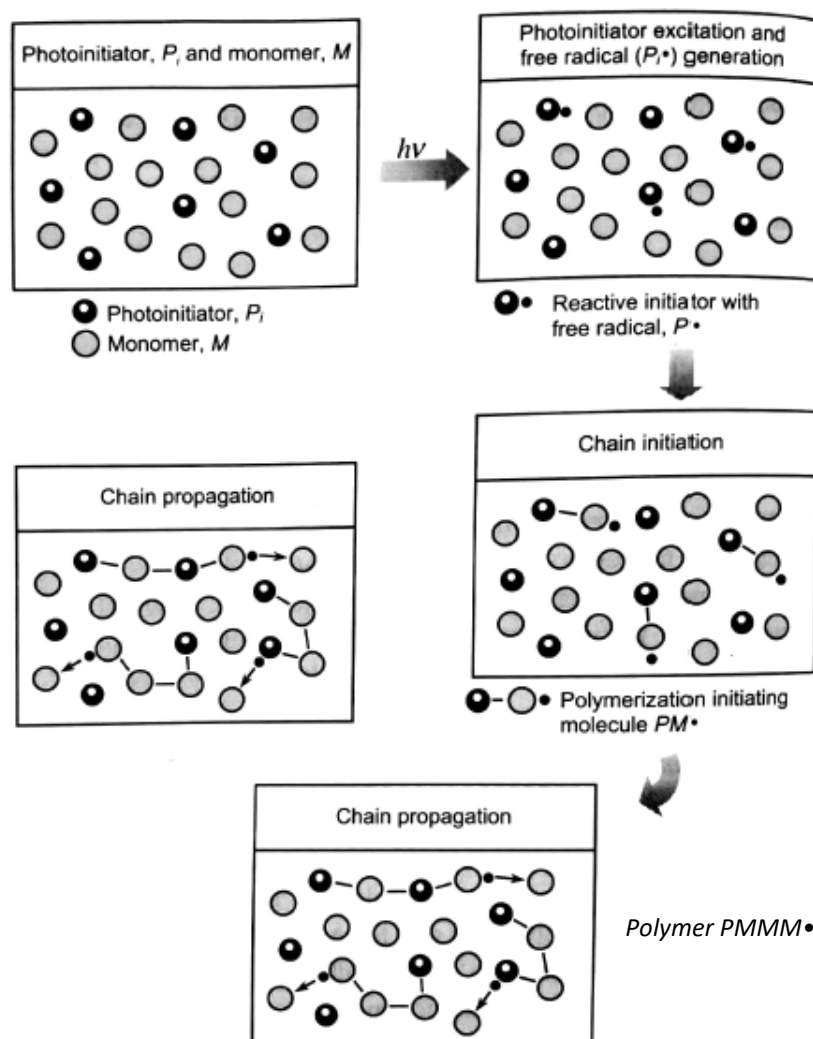
- Stereolithography Apparatus (SLA) – Uses a UV laser to selectively cure liquid photopolymer resin layer by layer.
- Digital Light Processing (DLP) – Employs a digital light projector to cure entire resin layers at once, enabling faster builds.
- Lithography-based Metal Manufacturing (LMM) – Utilizes photopolymer resin mixed with metal particles, later sintered to form dense metal parts.
- Light Enabled Additive Production (LEAP) – A high-speed process that cures resin using advanced light projection for rapid prototyping.

- Projection Micro Stereolithography (PμSL) – Provides microscale resolution using projected light patterns for fabricating microstructures.
- Masked Stereolithography (MSLA) – Uses an LCD mask with UV light to simultaneously cure entire resin layers, combining high speed and low cost.

In the present chapter will focus only on SLA process in detail with respect to its working principle, advantages, limitations, materials used and applications.

VAT POLYMERIZATION

Polymerization is the process of linking small molecules (known as monomers) into chain-like larger molecules (known as polymers). When the chain-like polymers are linked further to one another, a cross-linked polymer is said to be formed.



VAT Polymerization Principle

Photo polymerization is polymerization initiated by a photochemical process whereby the starting point is usually the induction of energy from the radiation source. Photoinitiator molecules (P_i), which are mixed with the monomers, M , are exposed to a Laser source of actinic photons, with energy of $h\nu$. The photoinitiators absorb some of the photons and are in an excited state. Some of these are converted into reactive initiator molecules ($P^\bullet$),

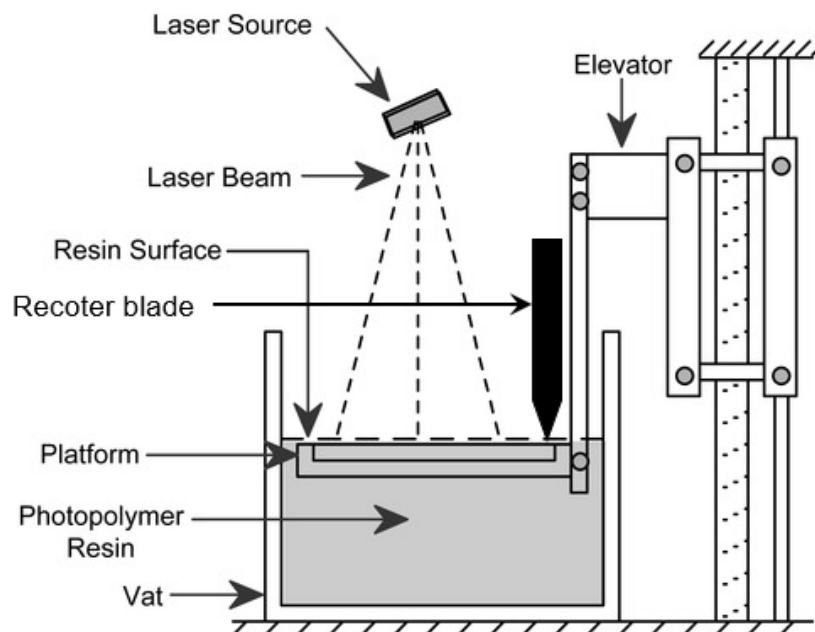
after undergoing several complex chemical energy transformation steps. These molecules then react with a monomer molecule to form a polymerization initiating molecule, (PM•). This is the chain initiation step. Once activated, additional monomer molecules go on to react in the chain propagation step, forming longer molecules, (PMMM•) until a chain inhibition process terminates the polymerization reaction. The longer the reaction is sustained, the higher will be the molecular weight of the resulting polymer.

1.2.2 STEREOLITHOGRAPHY (SLA)

3D Systems was founded in 1986 by inventor Charles W. Hull and entrepreneur Raymond S. Freed. Amongst all the commercial RP systems, the Stereolithography Apparatus (SLA) is the pioneer with its first commercial system marketed in 1988. By 2007 3D Systems is a global company that delivers advanced rapid prototyping solutions to every major market around the world.

SLA is a Liquid based RP process, parts are built from a photo-polymer resin. A photopolymer is a light-sensitive material that changes physical properties when exposed to light. Solidification happens from the impact of a laser light on the upper surface of the liquid by VAT polymerization.

HARDWARE



SLA Hardware and Principle

- Build Chamber: Contains a removable Vat that holds the build resin, a detachable, perforated build platen on a -Z axis elevator frame, and an automated resin-level checking apparatus. The build platen has a small amount of -Z movement capability, which allows the computer to maintain the exact height per layer.
- Recoater blade: A recoater blade rides along a track at the top of the vat and serves to "smooth" the liquid across the part surface to prevent a rounding of edges due to cohesion effects.

- Laser and Optics: The laser and optics required to cure the resin. The laser beam is transferred to the part surface below by a series of optics, to scan the cross section of the part being built.
- Post processing units: Include an alcohol bath and an Ultraviolet oven call the Post Curing Apparatus. Parts are washed in the alcohol or a similar solvent immediately after being removed from the machine. This step removes any extra resin that clings to the surfaces of the part. After the final supports are removed, with some build styles the parts are required to be placed in the Ultraviolet oven to finish fully curing.

PROCESS:

The process begins with the filament spools loaded into the FDM machine and the build platform positioned at the starting height. A three-dimensional CAD model, saved in STL format and sliced into layers, is loaded into the system. Any required support structures are generated to stabilize overhanging features during printing. The computer-controlled extrusion head then heats the filament to a semi-liquid state and deposits it along the tool path corresponding to the first layer on the build platform. After completing a layer, the build platform moves down by one layer height, and the extruder deposits the next layer of molten material. This layer-by-layer deposition continues from the bottom-up until the complete part is built. Once printing is complete, the part is removed from the build platform, and any support material is detached or dissolved to reveal the final component.

ADVANTAGES

1. SLA machines have **wide range of build volumes** from small to large to suit the needs of different users.
2. Parts printed **have good dimensional accuracy** within $\pm 0.1 - 0.2$ mm makes it ideal for precision prototyping.
3. SLA can obtain one of the **best surface finishes** amongst RP technologies.
4. There is a **wide range of resin materials**, from general-purpose materials to specialty materials for specific applications.
5. The process is of high resolution of 25–50 microns layer thickness allows **production of intricate geometries and build parts with complex details**.
6. SLA can produce **optically clear parts** for lenses, fluidic devices, or visual prototypes.

DISADVANTAGES

1. **Requires support structures** for printing parts that have overhangs and undercuts.
2. **Requires post-processing** for removal of supports and cleaning in solvents, which is tedious, time-consuming, and may damage the model.
3. **Requires post-curing** under UV light to achieve full strength and stability of the printed part.
4. Resin materials are generally **brittle and not suitable for high load-bearing** applications.
5. **Operational costs are relatively high** due to expensive resins, resin tanks, and maintenance.
6. **Build speed is slower** compared to some other AM processes, especially for large parts.

1.2.3 APPLICATIONS OF SLA

SLA is one of the most precise additive manufacturing technologies, well known for its ability to produce high-resolution parts with smooth surface finish using liquid photopolymer resins. Its unique features such as excellent accuracy, fine detailing, and ability to create complex geometries make it highly suitable for applications where precision and aesthetics are of prime importance. Following are the potential areas of application:

- **Concept Models** - Realistic appearance prototypes for product design validation.
- **Dental & Medical Models** - Crowns, surgical guides, anatomical models.
- **Jewellery Prototyping** - Master patterns for investment casting.
- **Miniatures & Figurines** - High-resolution prototypes with excellent aesthetics.
- **Optics Prototyping** - Lenses, casings, and light guides.

1.2.4 SLA MATERIALS

SLA resins are generally UV-curable photopolymers, based on acrylates, epoxies, urethanes, or composites with fillers (glass, ceramic, wax, carbon). Each formulation tailors properties for specific applications.

SLA Materials – Composition, Properties, and Applications

Category	Material Type	Key Properties	Typical Applications
<i>Standard Resins</i>	Photopolymer acrylates/epoxies	Smooth surface, high detail, brittle under stress	Concept models, visual prototypes
<i>Engineering Resins</i>	Urethane-based resins	Impact-resistant, ABS/PP-like strength	Functional prototypes, snap-fits
	Glass/ceramic-filled photopolymers	High stiffness, dimensional stability	Jigs, fixtures, precision parts
	Heat-resistant photopolymers	Withstands high heat, thermal stability	Mold masters, heat-resistant parts
	Urethane/elastic photopolymers	Rubber-like, bendable, resilient	Gaskets, seals, wearables
<i>Specialty Resins</i>	Wax-filled or castable acrylates	Clean burnout, no ash residue	Jewelry, dental crowns, investment casting
	Clear methacrylate resins	Optical clarity, smooth finish	Lenses, light pipes, fluidic devices
	Biocompatible methacrylate/epoxy blends	Safe for medical/dental use, sterilizable	Surgical guides, prosthetics
	Ceramic-filled resins	Very rigid, good thermal and dimensional stability	Industrial models, engineering prototypes
<i>Dental/Medical Resins</i>	Methacrylate-based	High precision, tooth-colored	Dental aligners, models
	Biocompatible methacrylate resins	Sterilizable, safe for patient contact	Surgical navigation aids